

Question 2 (20 Marks)

2.1 Consider the following Java program:

01	//Program 1
02	public enum Color {
03	RED("FF0000"), GREEN("00FF00"), BLUE("0000FF");
04	
05	private String hexCode;
06	
07	private Color(String hexCode) {
08	this.hexCode = hexCode;
09	}
10	
11	public String getHexCode() {
12	return hexCode;
13	}
14	public static void main(String [] a) {
15	for (Color color : Color.values())
16	System.out.println((color.ordinal() + 1) + ") " + color +
17	" has code " + color.getHexCode());
18	}
19	}
20	

Based on the program above that defines an enumerated type called **Color**, answer the following questions:

a) What is the significance of the values "FF0000", "00FF00", and "0000FF" that are passed as arguments to the constructor of each value in the **Color** enumerated type?

(1 mark)

b) What is the purpose of the **getHexCode()** method in the **Color** enumerated type?

(1 mark)

c) What will be the output of the program?

(3 marks)

d) Write a Java statement in **main()** method that uses the **compareTo()** method to display the results of a comparison of the positions of **RED** and **BLUE** in the **Color** enumerated type. What will the output be?

(2 marks)

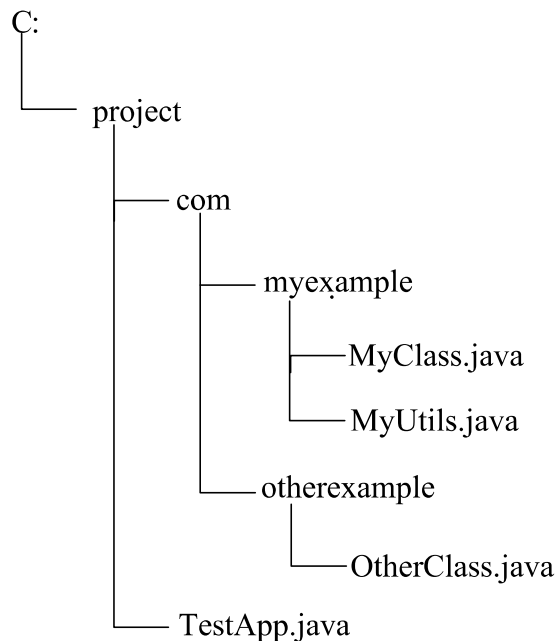
e) Consider adding the following statements in the **main()** method:

```
if (Color.GREEN.equals(Color.RED))
    System.out.println("GREEN and RED are two different colors.");
else
    System.out.println("GREEN and RED are the same color.");
```

The statements, however, include an error. What is the error, and how can it be fixed?

(2 marks)

2.2 Given the directory structure as shown below:



a) What is the package name of **MyClass.java** and **MyUtils.java**?

(1 mark)

b) What is the package name of **OtherClass.java**?

(1 mark)

c) In **TestApp.java**, how would you import all the classes in **com.myexample**?

(1 mark)

d) Suppose you want to move **MyUtils.java** to a new package called **com.myexample.util**. What changes would you need to make to the directory structure and the code?

(3 marks)

2.3 Consider the Java program below and answer the following questions:

```
01 //Program 2
02 public class WrapperClass {
03     public static void main(String[] args) {
04         Integer num1 = 10;
05         Double num2 = 3.14;
06
07         //(a)Converting an object of a wrapper type to
08         //its corresponding primitive value
09         int value1 = _____;
10         double value2 = _____;
11
12         System.out.println("value1: " + value1);
13         System.out.println("value2: " + value2);
14
15         //(b)Adding value1 and value2 and storing the
16         //result in the result variable by converting
17         //it to the Double wrapper class
18         Double result = _____;
19
20         System.out.println("result: " + result);
21     }
22 }
```

- a) Fill in the blanks with an appropriate Java statement to convert an object of a wrapper type to its corresponding primitive value. **Note:** you must use a method provided in the wrapper class. What is the name of the conversion?

(3 marks)

- b) Fill in the blanks with an appropriate Java statement to add **value1** and **value2** and store the result in the **result** variable by converting it to the **Double** wrapper class. **Note:** you must use a method provided in the wrapper class.

(2 marks)